Downscaling Municipal Soybean Yield Labels to Pixel Predictions with MSTNet in Paraná, Brazil

Abstract

Official crop productivity is often available only for coarse administrative units, such as municipalities in Brazil and similar units in many other countries. Monitoring and insurance, however, frequently need estimates at the field scale or for other sub-municipal regions. We propose MSTNet (*Multi* STNet), a Transformer model that combines data sampled at different temporal frequencies. Separate STNet trunks encode soy-masked Sentinel-2 time series and a daily gridded climate series, while annual soil properties enter the decoder by late fusion. The model predicts one value per soy pixel. Municipal PAM labels supervise the mean of those pixel predictions, so the trained weights can be aggregated to arbitrary polygons at inference without retraining. We evaluate the approach in Paraná, Brazil, on harvest years 2020–2024, with municipalities from 2021 held out for validation and test. On the test municipalities, the municipal aggregates reach $R^2 = 0.398$. Direct comparison with earlier work is difficult because few studies train under municipal yield labels while scoring pixels for flexible downscaling. We therefore report a clear holdout benchmark and leave multi-year and field-referenced checks for future work.

1 Introduction

Reliable soybean productivity estimates matter for agricultural monitoring, insurance, and risk assessment. In Brazil, the Municipal Agricultural Production survey (PAM) from the Brazilian Institute of Geography and Statistics (IBGE) publishes official yields only at the municipality level. The same kind of coarse reporting is common elsewhere. Sentinel-2 imagery, gridded weather, and soil maps describe crop condition at much finer scales, but field-level yield labels are rarely available statewide. What we need is a model that can learn from municipal labels and still make predictions for smaller regions, at a scale that is cheap enough to run over large areas.

MSTNet (*Multi* STNet) is our answer to that requirement. It extends the Spectral–Temporal Network (STNet) of Xu et al. [2024] by giving each covariate stream its own trunk when the streams are sampled at different frequencies. Here, one trunk encodes soy-masked Sentinel-2 reflectance and another encodes daily climate. Annual MapBiomas Solo properties, which are static once the harvest year is fixed, are concatenated in the decoder. Predictions are defined per soy pixel. For speed, training fits the network on municipality-mean Sentinel-2 series supervised by PAM. At evaluation and inference we apply the same weights to every soy pixel and average the scores to municipalities (or to any other polygon) without retraining.

We run the experiments on soybean municipalities in Paraná for harvest years 2020–2024 and hold out 2021 for validation and test. Because PAM is the only supervised target, the numbers we report are municipal: holdout R^2, RMSE, and related diagnostics on municipal aggregates. We are not aware of many yield studies that train a multi-frequency Transformer on municipal labels while still exposing pixel scores for downscaling. Matched baselines for that setup are scarce, so we present the municipal holdout as a reproducible benchmark rather than a direct comparison against the literature.

Our contributions are MSTNet itself, the pixel-scoring setup that turns municipal supervision into sub-municipal maps, and a fixed 2021 municipality holdout for Paraná over 2020–2024.

Table 1: Summary of data sources used in the experiments.

Category	Variables and processing	Source
Satellite	Ten Sentinel-2 L2A reflectance bands (B2, B3, B4, B5, B6, B7, B8, B8A, B11, B12), soy-masked zonal means via Google Earth Engine	Copernicus Sentinel-2 L2A [Copernicus Sentinel-2, processed by ESA, 2021]; GEE [Gorelick et al., 2017]
Climate	Daily precipitation, ETo, Rs, Tmax, and Tmin; averaged across Xavier cells within the municipality polygon	Xavier BR-DWGD [Xavier et al., 2022]
Soil	Clay, silt, sand, and soil organic-carbon stock at 0–30 cm; late-fused static covariates	MapBiomas Solo Collection 3 [MapBiomas, 2025b]
Crop mask	MapBiomas soybean class 39, used to select pixels before Sentinel-2 aggregation	MapBiomas Collection 10 [MapBiomas, 2025a]
Target	Municipal soybean productivity (t/ha)	IBGE PAM [Instituto Brasileiro de Geografia e Estatística, 2026]

2 Data

The analysis uses municipal PAM productivity labels together with Sentinel-2, daily climate, and soil covariates (Table 1). Soy-masked Sentinel-2 pixels provide the spectral signal. Climate is reduced to a municipal daily series. Soil can be attached per pixel or as a municipal vector. Figure 1 shows the municipality-mean spectral path used for training. Appendix A shows the pixel-by-pixel path used for the reported holdout metrics and for sub-municipal maps.

2.1 Study area and labels

The study covers soybean-producing municipalities in the state of Paraná, Brazil. The target is municipal soybean productivity in tonnes per hectare (t/ha) from IBGE PAM [Instituto Brasileiro de Geografia e Estatística, 2026]; municipality code and harvest year identify each observation. No field- or pixel-level yield labels are used. We retain municipality–year pairs whose Sentinel-2 coverage ratio relative to the reported planted area lies in $[0.8, 1.2]$, so that satellite coverage is roughly consistent with official area statistics.

2.2 Covariates and municipal aggregation

Sentinel-2 L2A reflectance is processed in Google Earth Engine [Gorelick et al., 2017]: for each municipality and acquisition date we compute soy-masked zonal means of ten bands (B2, B3, B4, B5, B6, B7, B8, B8A, B11, B12), using MapBiomas Collection 10 class 39 as the soybean mask [MapBiomas, 2025a]. Each municipal spectral sequence carries a day-of-year index for positional encoding and is truncated or padded to $T = 45$ acquisitions. Climate covariates come from the Xavier daily weather grid [Xavier et al., 2022] as a separate municipal daily series over the inclusive Oct 1–Mar 31 window (length 183): precipitation, reference evapotranspiration (ETo), solar radiation (Rs), maximum temperature (Tmax), and minimum temperature (Tmin). Daily climate is averaged over Xavier cells whose centers fall in the municipality polygon and is scaled independently of reflectance; MSTNet encodes it in a dedicated trunk rather than concatenating climate onto Sentinel-2 timesteps. Static MapBiomas Solo properties (clay, silt, sand, and SOC stock at 0–30 cm) are late-fused in the decoder [MapBiomas, 2025b]. An ablation omits climate and soil inputs and uses the ten spectral bands alone.

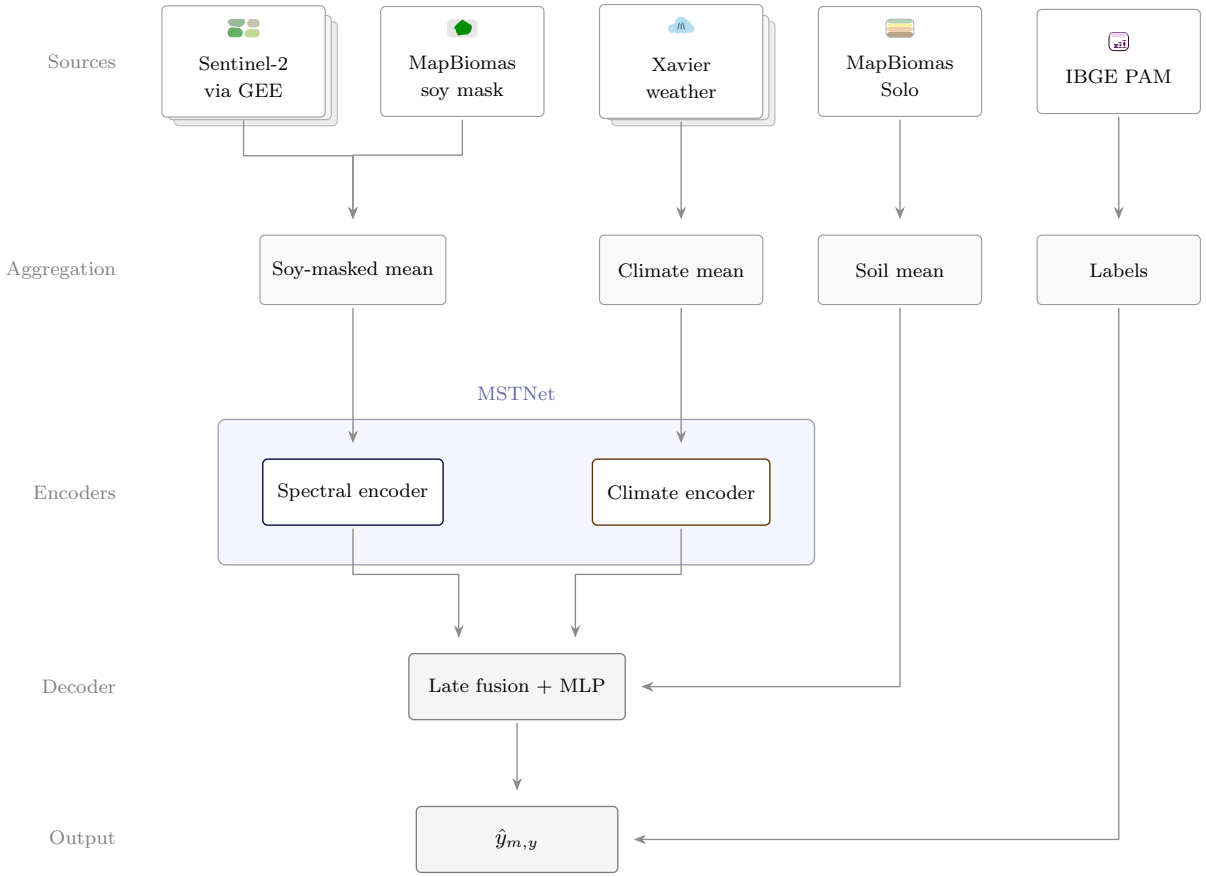


Figure 1: Data pipeline for MSTNet. Sentinel-2 (soy-masked) and Xavier weather are encoded by parallel trunks; MapBiomas Solo enters the decoder by late fusion. The decoder prediction $\hat{y}_{m,y}$ is supervised by IBGE PAM labels at the output.

2.3 Train / validation / test split

Splits follow a year-holdout design on municipalities. All municipality–year observations outside the holdout year are used for training; municipalities in the holdout year are partitioned into validation and test sets. The holdout year is **2021**. After restricting to harvest years $\{2020, 2021, 2022, 2023, 2024\}$, the training set contains 997 municipality–years, while validation and test contain 132 and 123 municipalities in 2021, respectively. Training target statistics used for loss normalization are mean 3.17 t/ha and standard deviation 0.93 t/ha. All reported metrics are municipal aggregates obtained by averaging pixel-level scores (Section 3.1). We do not validate pixel maps against field yields.

3 Methods

3.1 Pixel scoring under municipal supervision

Let $\mathcal{P}_{m,y}$ denote the soybean pixels observed in municipality m in harvest year y. MSTNet defines a productivity score $\hat{y}_p$ for each soy pixel p from its Sentinel-2 sequence $\mathbf{x}_p$, the municipal daily climate series $\mathbf{c}_{m,y}$ (broadcast to all pixels in m), and soil covariates $\mathbf{s}_p$ (or a municipal soil vector when only polygon-level soil is available). Reported municipal predictions average those pixel scores,

$$\hat{y}_{m,y} = \frac{1}{|\mathcal{P}_{m,y}|} \sum_{p \in \mathcal{P}_{m,y}} \hat{y}_p, \qquad \hat{y}_p = f_\theta(\mathbf{x}_p, \mathbf{c}_{m,y}, \mathbf{s}_p), \tag{1}$$

and are compared with the PAM label $y_{m,y}$. At inference the field $\{\hat{y}_p\}$ can also be kept as a map or averaged over any sub-municipal polygon. Appendix A sketches this pipeline.

Training uses a cheaper surrogate: soy-masked Sentinel-2 reflectance is averaged before encoding,

$$\bar{\mathbf{x}}_{m,y,t} = \frac{1}{|\mathcal{P}_{m,y,t}|} \sum_{p \in \mathcal{P}_{m,y,t}} \mathbf{x}_{p,t}, \tag{2}$$

so each municipality–year is one forward pass $\hat{y}^{\text{train}}_{m,y} = f_\theta(\bar{\mathbf{x}}_{m,y}, \mathbf{c}_{m,y}, \mathbf{s}_m)$ supervised by PAM. The holdout metrics in Sections 4–5 are not from that surrogate: they apply the trained weights pixel by pixel and aggregate with (1). Because f_θ is nonlinear, mean-then-encode and encode-then-mean need not coincide; the pixel-mean evaluation is the operational downscaling check.

3.2 Architecture

MSTNet (*Multi* STNet) places one STNet trunk on each time-varying stream so that modalities with different sampling rates need not share a common calendar [Xu et al., 2024]. A spectral trunk encodes Sentinel-2 sequences (ten bands, length $T = 45$) with day-of-year positional encodings and NDVI-weighted temporal pooling. A climate trunk encodes daily Xavier weather (five channels: precipitation, ETo, Rs, Tmax, Tmin) over the Oct–Mar window (length 183), pooled by a mean over valid days. MapBiomas Solo properties are fixed once the harvest year is known. They are concatenated with the pooled trunk embeddings and passed through a multilayer-perceptron decoder that outputs a productivity estimate (per pixel or per municipal aggregate, depending on the input path). As an ablation on a spectral-only single-trunk STNet, we also try mean pooling over valid spectral timesteps instead of NDVI weights. That spectral-only model uses ten-band inputs and has no climate or soil branches.

Each MSTNet trunk uses one Transformer layer, $d_{\text{model}} = 128$, two attention heads, feed-forward width $d_{\text{inner}} = 256$, and dropout 0.4.

3.3 Loss and optimization

Training minimizes mean squared error between predicted and observed municipal productivity after standardizing targets with the training-set mean and standard deviation. Reported metrics (RMSE, MAE, and R^2) are computed after transforming predictions back to t/ha.

Optimization uses AdamW with learning rate 1×10^{-3}, weight decay 5×10^{-4}, and batches of 40 municipalities. The learning rate is scheduled to drop at epochs 40 and 120; training stops early after fifteen epochs without improvement in validation R^2. No self-supervised pretraining is used: the network is trained from scratch under municipal supervision.

4 Experiments

4.1 Proposed model

The proposed model is MSTNet: one STNet trunk for Sentinel-2, one for daily Xavier climate (precipitation, ETo, Rs, Tmax, and Tmin), and late fusion of MapBiomas Solo properties in the decoder. Each trunk uses one Transformer layer with $d_{\text{model}} = 128$, two attention heads, $d_{\text{inner}} = 256$, and dropout 0.4. Reported metrics apply the selected weights pixel by pixel and average the scores to municipalities (Section 3.1). We select the epoch with the highest validation R^2 of those municipal aggregates; that epoch is 12. The selected checkpoint is evaluated once on the held-out test municipalities.

Table 2: Validation and test metrics for the MSTNet municipal predictor at the validation-selected epoch (12). RMSE and MAE are in t/ha.

Split	RMSE	MAE	R^2
Validation (2021)	0.313	0.242	0.424
Test (2021)	0.309	0.237	0.398

4.2 Spectral-only reference

For context, we report results from a separate grid search using spectral inputs only: $d_{\text{model}} \in \{64, 96, 128\}$, number of heads $\in \{4, 8\}$, $d_{\text{inner}} \in \{64, 128\}$, and dropout $\in \{0.3, 0.4\}$, with a single Transformer layer. Twenty-four configurations completed successfully. The best spectral-only model ($d_{\text{model}} = 96$, eight heads, $d_{\text{inner}} = 64$, dropout 0.3) is reported as a reference. It differs from MSTNet in both inputs and architecture, so this is not a controlled estimate of what climate or soil add on their own.

4.3 Evaluation protocol

Held-out evaluation compares municipal predictions with PAM labels on the 2021 validation and test municipalities. We also report learning curves, predicted-versus-observed municipal scatterplots for harvest years 2020–2024, and a municipal error map for the 2021 holdout.

4.4 Ablations

We report two auxiliary comparisons: the spectral-only reference above, and a temporal-pooling comparison that swaps NDVI-weighted pooling for mean pooling over valid timesteps in the spectral-only architecture. These are descriptive checks, not controlled multimodal ablations.

5 Results

5.1 Municipal prediction accuracy

Table 2 summarizes validation and test metrics for the proposed MSTNet municipal predictor. Validation R^2 peaks at epoch 12 (0.424), with RMSE 0.313 t/ha. The corresponding test metrics are $R^2 = 0.398$ and RMSE 0.309 t/ha. The validation and test results are similar, with errors near 0.31 t/ha on the fixed 2021 holdout split.

Figure 2 shows learning dynamics. Validation RMSE and R^2 improve through epoch 12. After that, validation fluctuates (no learning-rate drop falls in this window), and training stops early at epoch 27 after fifteen epochs without a new validation R^2 best.

5.2 Spectral-only reference

Table 3 lists the five strongest configurations in the separate spectral-only search. We include them to document the reference run, not to isolate climate or soil effects, since the architecture differs from MSTNet.

5.3 Auxiliary comparisons

Spectral-only reference. The top spectral-only configuration achieves validation $R^2 = 0.165$ (RMSE 0.376 t/ha) and test $R^2 = 0.117$ (RMSE 0.375 t/ha). MSTNet is stronger on this split, but the experiments also differ in architecture and optimization, not only in which covariates

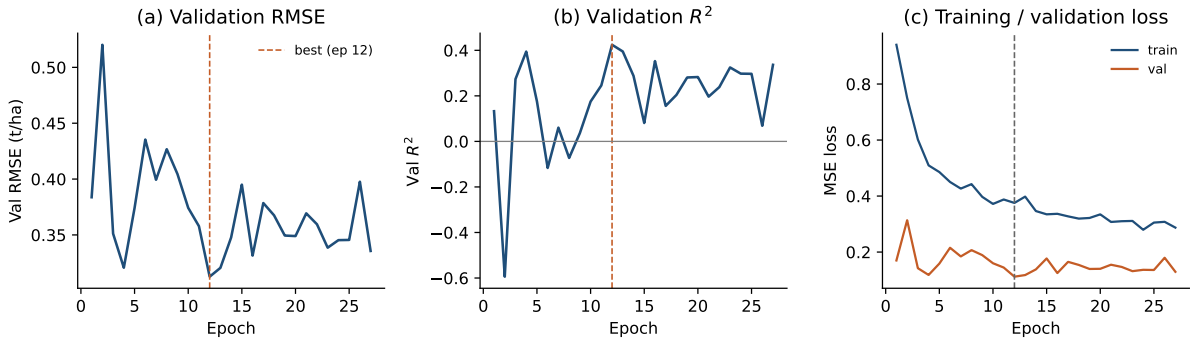


Figure 2: Training dynamics for MSTNet. The dashed line marks the validation-selected epoch (12).

Table 3: Top five configurations from the separate spectral-only search (24 completed runs), ranked by validation R^2. This is not a matched ablation of MSTNet.

Rank	d_{model}	Heads	d_{inner}	Dropout	Val R^2	Test R^2
1	96	8	64	0.3	0.165	0.117
2	96	4	128	0.4	0.084	0.134
3	64	4	64	0.4	0.076	—
4	64	4	128	0.3	0.071	0.119
5	64	4	128	0.4	0.046	0.106

are available. Attributing the gap to climate or soil would need a controlled modality ablation with the same network and training protocol.

Temporal pooling. With the best spectral-only architecture fixed, replacing NDVI-weighted temporal pooling by mean pooling over valid timesteps reduces validation R^2 from 0.165 to 0.049 (best epoch 12), while test R^2 remains nearly unchanged (0.117). NDVI-weighted pooling improves validation fit in this spectral-only comparison, but this improvement does not translate into a clear test gain on the single holdout year.

5.4 Per-year municipal diagnostics

Figure 3 shows predicted versus observed municipal productivity by harvest year. The 2021 value ($R^2 = 0.395$) agrees closely with the checkpoint's held-out test metric. In 2022, R^2 is 0.612 but RMSE increases to 0.709 t/ha. This combination is possible because R^2 measures relative variation around that year's mean, while RMSE measures errors in the original productivity scale. Results for 2023–2024 are intermediate (Table 4).

Figure 4 maps municipal prediction error in the 2021 holdout year. Figure 5 shows an intra-municipal example: pixel-level productivity scores for Guarapuava under the same checkpoint. That map is qualitative; we have no field yields to score it against, only the municipal PAM total when the pixels are averaged.

6 Discussion

On the 2021 municipality holdout, validation and test stay close: RMSE is about 0.31 t/ha, with $R^2 = 0.424$ on validation and $R^2 = 0.398$ on test. That agreement suggests the selected epoch did not overfit the validation partition hard, but it says little about other years.

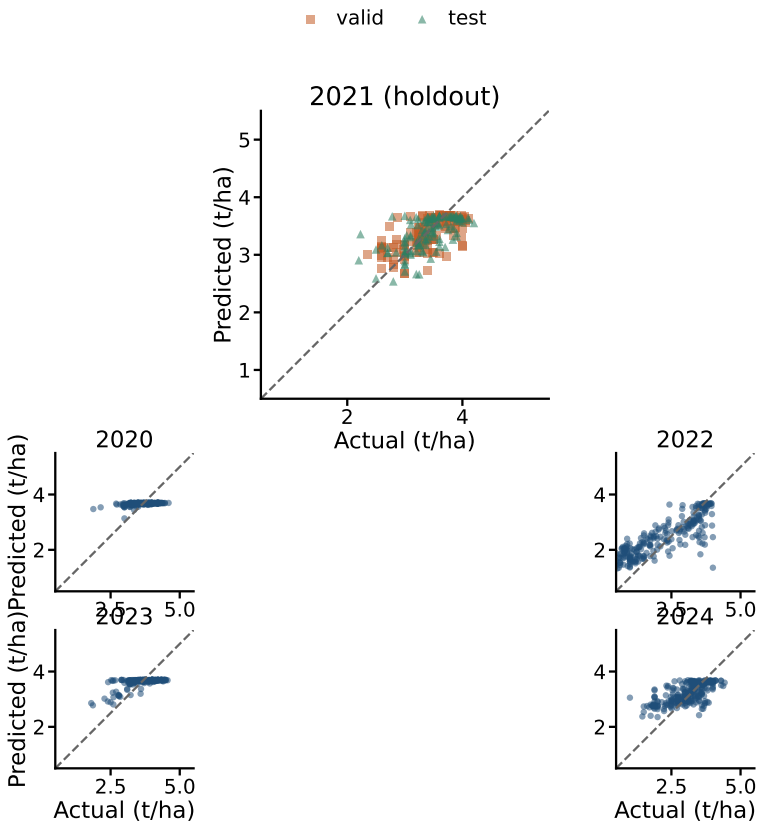


Figure 3: Predicted versus observed municipal soybean productivity (t/ha) for MSTNet. Top: holdout year 2021 (validation and test municipalities). Bottom: remaining harvest years in a 2×2 panel. Points are colored by split. The dashed line is identity.

MSTNet beats the separately tuned spectral-only reference on this split. The gap is interesting, yet we cannot pin it on climate or soil alone: the two experiments also differ in architecture and optimization. A fairer ablation would freeze those choices and add climate and soil one at a time.

The spectral-only search preferred a small Transformer, while MSTNet uses a larger two-trunk design. Together with the validation swings after epoch 12, that points to capacity and optimization as first-order factors. Repeated seeds or a nested selection protocol would make that sensitivity clearer.

The 2022 numbers show why R^2 alone is not enough. Cross-municipality R^2 reaches 0.612, but RMSE climbs to 0.709 t/ha, well above the 2021 level. Operational reports should keep both absolute error and explained variance, and they should call out drought years explicitly.

In the spectral-only architecture, NDVI-weighted temporal pooling raised validation R^2 relative to mean pooling, while test R^2 barely moved. That looks more like a model-selection effect than a reliable generalization gain.

Because scores are defined per pixel, municipal PAM labels can in principle support downscaling to arbitrary sub-municipal shapes (Figure 5). The numbers in this paper evaluate municipal aggregates of those pixel scores; they do not check intramunicipal maps against field yields. Training itself used municipality-mean Sentinel-2 series for speed, so the reported skill is a transfer check of those weights under encode-then-mean evaluation. We also have only one holdout year, no repeated runs, few matched external baselines for this training setup, and clear sensitivity to unusual seasons. Next steps are multi-year and geographic holdouts, controlled ablations, stronger baselines, and independent yield data when the goal is spatially resolved accuracy.

Table 4: Per-year municipal prediction metrics for MSTNet.

Year	n	RMSE	MAE	R^2
2020	249	0.384	0.313	0.101
2021	255	0.316	0.244	0.395
2022	240	0.709	0.568	0.612
2023	248	0.405	0.322	0.304
2024	260	0.531	0.409	0.374

7 Conclusion

We presented MSTNet, a multi-frequency Transformer that uses Sentinel-2 time series, daily Xavier climate, and MapBiomas Solo covariates to predict soybean productivity from municipal PAM labels while scoring soy pixels for sub-municipal maps. On the 2021 municipality holdout for Paraná (harvest years 2020–2024), municipal aggregates at the selected epoch reach $R^2 = 0.424$ and RMSE $= 0.313\,\mathrm{t/ha}$ on validation, and $R^2 = 0.398$ and RMSE $= 0.309\,\mathrm{t/ha}$ on test.

These figures give a reproducible municipal benchmark for the setting we study. Calibrated intramunicipal maps and claims about the value of each covariate group still need field checks, matched ablations, and multi-year holdouts.

Data availability

Municipal productivity labels are from IBGE PAM and provide the only supervised signal. Sentinel-2 L2A reflectance was processed in Google Earth Engine with MapBiomas soybean masking; daily Xavier climate and MapBiomas Solo enter as described in Section 2. Training uses municipality-mean Sentinel-2 series; reported evaluation scores soy pixels and averages to municipalities (Sections 3 and A). Further details are available from the authors upon reasonable request.

A Pixel-level MSTNet pipeline

Section 3.1 defines the pixel-scoring evaluation (1): soy-masked Sentinel-2 pixels are scored one by one, municipal climate is shared across pixels, soil enters by late fusion, and the mean of the pixel predictions is compared with PAM. Figure 1 shows the municipality-mean spectral path used for training. Figure 6 shows the pixel-by-pixel path used for the reported holdout metrics and for intramunicipal maps. At inference the pixel field $\{\hat{y}_p\}$ can be kept as a map or averaged over any polygon. That map is not a training target.

References

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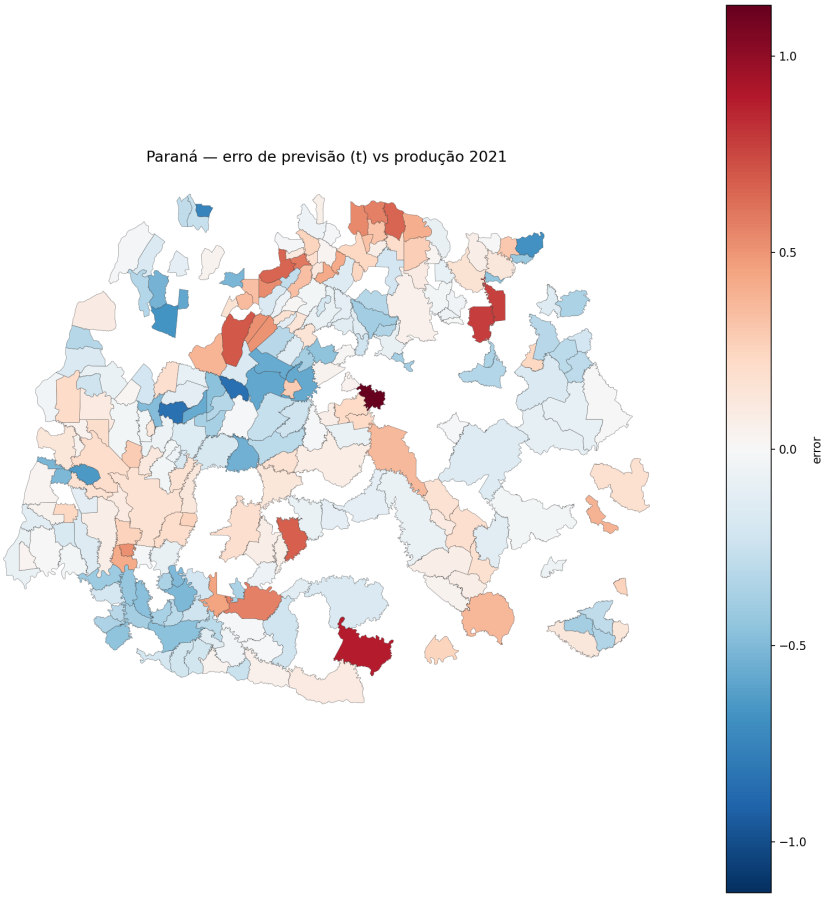


Figure 4: Municipal aggregate prediction error map for harvest year 2021 (holdout).

MapBiomas. Collection 10 of the annual land cover and land use maps of brazil (1985–2024), 2025a.

MapBiomas. Soil particle size and texture maps for brazil 0–100 cm: Mapbiomas soil collection 3 (beta), 2025b.

Alexandre Cândido Xavier, Bridget R. Scanlon, Carey W. King, and Aline I. Alves. New improved Brazilian daily weather gridded data (1961–2020). *International Journal of Climatology*, 42 (16):8390–8404, 2022. doi: 10.1002/joc.7731.

Yijia Xu, Yuchi Ma, and Zhou Zhang. Self-supervised pre-training for large-scale crop mapping using Sentinel-2 time series. *ISPRS Journal of Photogrammetry and Remote Sensing*, 207: 312–325, 2024. doi: 10.1016/j.isprsjprs.2023.12.005.

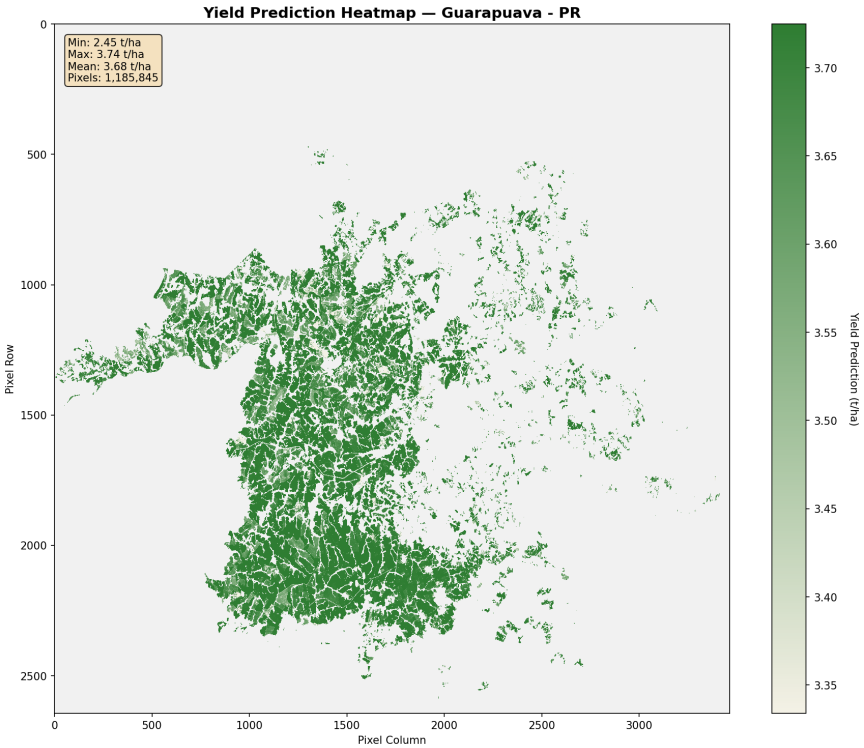


Figure 5: Intramunicipal map of downscaled pixel-level productivity predictions for Guarapuava, Paraná (holdout year 2021).

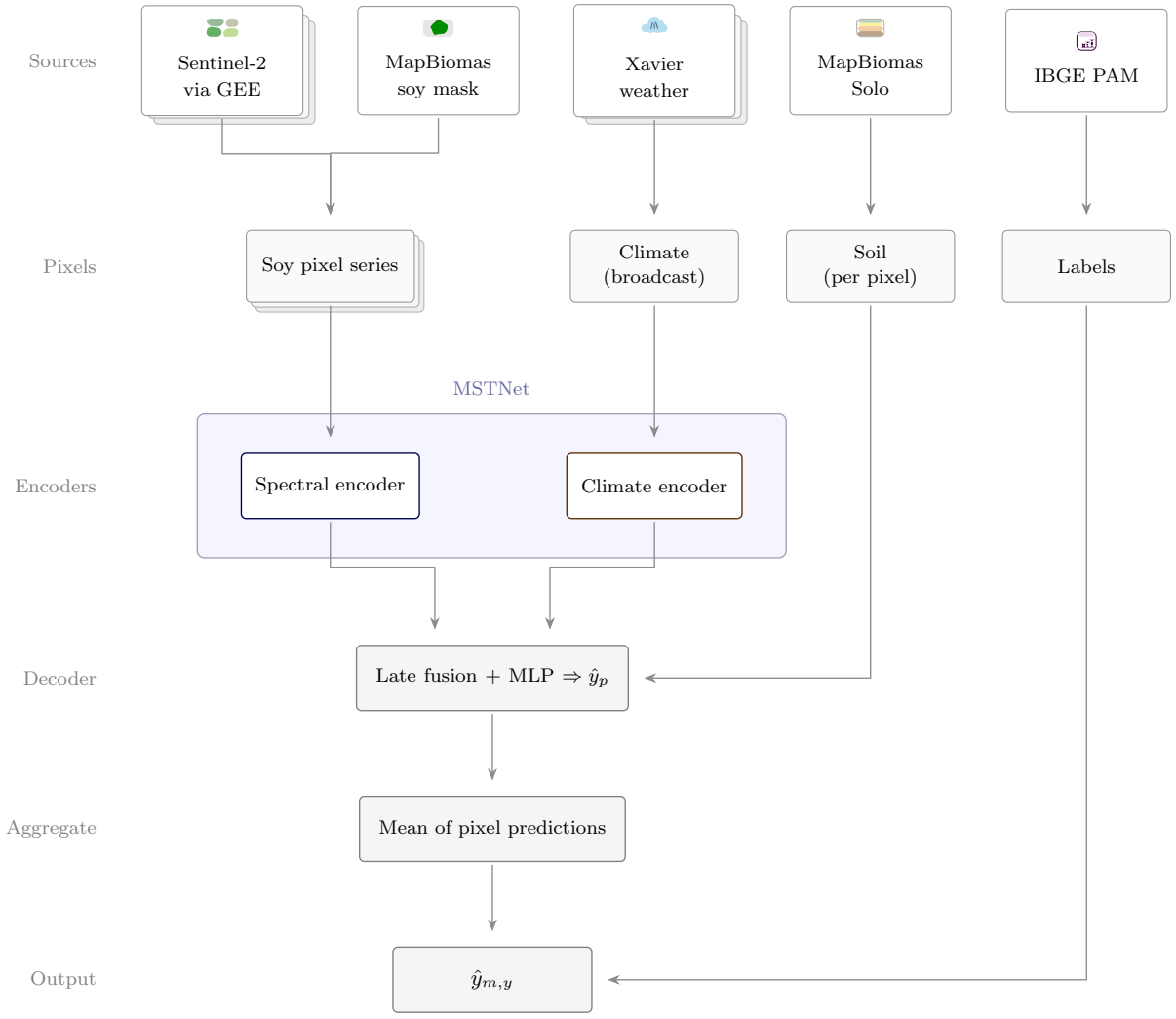


Figure 6: Pixel-by-pixel MSTNet pipeline. Soy-masked Sentinel-2 pixels are encoded individually; municipal Xavier climate is broadcast to pixels and soil enters by late fusion. The mean of pixel predictions $\hat{y}_p$ yields $\hat{y}_{m,y}$, which is supervised by IBGE PAM.